

Chapter 65 — Adolescent and Transitional Urology

WHAT CONSTITUTES ADOLESCENT UROLOGY?

- **Origin:** the recognition that patients born with complex urogenital conditions need **long-term adult care**.
- **Evolution:** from a stepwise paediatric → adolescent → adult sequence to **seamless lifelong care**.
Adolescent urology is the **bridge** between paediatric and adult urology.
- **Adult reconstructive urologists can perform some of the operations but are underequipped to care for the whole patient**, who brings:
 - **multiple previous operations;**
 - an **anatomical, psychological and physiological landscape altered** by the condition and prior surgery;
 - lifelong contact with health care, so patient and parents are often **expert in the condition** and unforgiving of a provider who is not;
 - expectations formed in a **paediatric hospital**, which must be **managed** for the adult relationship to succeed.
- **Many are underserved in adulthood.** Clinical and inter-centre collaboration is needed to publish **meaningful outcome data**.
- **Role of the adolescent urologist:** inform and recruit adult urologists into the field; publish and discuss outcomes with paediatric colleagues.

Concept — *The gap is in **whole-patient knowledge, not operative skill**. Childhood reconstruction creates an adult whose anatomy, history and expectations differ from those of any other adult patient.*

DEFINITION OF TRANSITION

- **Got Transition definition:** the move from a **child/family-centred to an adult/patient-centred model of care**, with or without a change of clinician. It has three parts: **planning → transfer → integration**.
- **Goals:** (1) improve young people's ability to **manage their own health care** and use services effectively; (2) an **organised process in both paediatric and adult practices** covering preparation, transfer and integration.
- **In practice:** **independence** in the care environment, **education** about the condition and its treatment, and **navigation** of adult care.
- **Transition vs transfer:** transition is the **whole process**; transfer is **one step within it** and need not involve a new clinician.

WHY IS TRANSITION NEEDED?

- **Agency for Healthcare Research and Quality (AHRQ) report:** on first reading, transition care in its present form **does not appear to make a significant difference**.
- **But robust, reproducible metrics are lacking.** Renal function and glycated haemoglobin (HbA1c) are expected to **worsen with age**, so **stability is success**, not improvement. **Better metrics** are needed.
- **Consistent finding:** **fewer patients are lost to follow-up** when actively transitioned to a capable adult service.

- **Avoid the word "normal":** the **patient is their own baseline**. Comparing an adult born with exstrophy to "normal" is meaningless and may stigmatise. Acknowledge the patient's wish for normality with sensitivity in order to gain trust.

Concept — *The evidence for transition looks weak because **the wrong endpoint is being measured**. In progressive congenital disease, **stability and retention in follow-up are the outcome**.*

WHO NEEDS TRANSITION?

- Congenital and childhood-acquired conditions both carry risk from the **condition or its prior treatment**, so **most patients need adult care**.

CATEGORIES REFERRED TO A UROLOGICAL TRANSITION SERVICE

Category	Examples	Implication
1. Congenital, complex needs; planned monitoring essential	Exstrophy, posterior urethral valves, complex hypospadias, spina bifida (SB)	Planned lifelong care
2. Treated congenital condition, now quiescent or mild , with risk through puberty	Vesicoureteric reflux (VUR)	May be discharged after an early-adult consultation, with open return
3. Significant childhood condition cured , with treatment sequelae	Pelvic rhabdomyosarcoma	Surveillance for late effects of treatment
4. New condition or concern beyond paediatric scope	Fertility, pregnancy	Needs adult expertise

- **Trap:** not every transitioned patient needs lifelong follow-up (category 2).
- **Purpose of successful transition:** reporting **long-term outcomes** for quality improvement, and **preventive care before catastrophic outcomes** to reduce costs.

WHAT HAPPENS WHEN TRANSITION FAILS?

- **Begin discussions in early adolescence**, individualised to the patient's developmental and physical needs.
- **Risks of non-transition:** a young adult who **does not understand their condition**; **overbearing family control**; **loss to follow-up**; **unrecognised complications**; **reliance on emergency care**; missing social and peer support.
- **Net result:** non-adherence, more emergency department (ED) attendances and hospitalisations, higher **morbidity and mortality**, and family dissatisfaction.

Concept — *Failed transition does its harm through **loss of preventive surveillance**. The patient reappears only through the emergency department, after complications that could have been prevented.*

TRANSITION AS A PROCESS

Constituent Elements of the Process

FIG. 65.1 AN OVERVIEW OF THE PROCESS: THE THREE PS OF TRANSITION

P	Content	Measure of success
Preparation	Supporting independence in the care environment. Can begin at any age, but usually starts in early adolescence (12–14 years) . Readiness tools such as the Transition Readiness Assessment Questionnaire (TRAQ) ; an electronic health record (EHR) tool may ease data collection	All patients and families shown to have been involved in preparation
Plan	Expected clinic schedule and frequency of imaging and blood tests . Self-care: catheterisation frequency, bladder irrigation, medications, when and whom to call	A plan for every patient, communicated to them and their future team
Provider of the future	May or may not be a new team; the patient knows which teams to see and where	A named future provider for every patient

- **The paediatric provider must initiate a formal transition process.** "Completing" care and leaving the patient to find their own way is unacceptable. The adult system must provide space, time and people.

Stakeholders and Their Roles

Patient

- The **primary stakeholder**. The patient must be medically and physically stable for handover, so **avoid transition during a crisis** such as deteriorating renal function or a need for urgent surgery.
- **Exception, pregnancy:** obstetric care lies outside a paediatric hospital, so transfer is **expedited**, with **co-management by paediatric and adult teams**. This keeps the familiar team involved and shows the teams working together.

Parents

- **Handing decision-making to their child** can overwhelm parents who have managed care since infancy: **pride, a need to show trust, and fear**. **Separating from a trusted provider** is itself a barrier.
- **Create private time for the patient alone**, especially for **sex, sexual function and sexual debut**. Then ask whether the patient wants the conversation shared with their parents. These conversations are **often omitted**; in some cohorts most patients had never discussed sex with a provider.

Health Care Providers

PAEDIATRIC VS ADULT SETTING

Paediatric	Adult
Multidisciplinary clinics common; several teams in one visit	Multidisciplinary care harder to establish
Cross-disciplinary learning between providers	Communication between specialists often lacking ; mitigated by complex-care schedulers and care coordinators

Health Care Institutions

- Access to adult care is difficult for young adults with complex chronic conditions. The conditions are **rare and heterogeneous**, so their financial benefit is **hard to quantify**. Show need through **practice growth**, and engage "**allies**": like-minded specialists, patient advocacy groups and the patients themselves.

When to Begin and Complete the Process

- **Begin:** the American Academy of Pediatrics (AAP) recommends about age 12. Content widens through the teens to cover diagnosis, past decisions, treatments and future considerations.
- **Practical skills:** booking appointments themselves, navigating insurance, knowing whom to call, and what to do in an emergency. **The most important element is teaching how and where to find valid information.** Written questions and a **portable health record** help.
- **Complete:** a paediatric setting becomes inappropriate for most patients by the **late teens to early 20s**. **Age 26** is the commonly cited latest age for completion, being the upper age for cover on a parent's insurance plan in the United States.
- **Where independent living is not achievable** (cognitive or physical limitation), the setting is a matter of judgement and needs ongoing reassessment. Plan **medical custodianship** (for example, a sibling as parents age or die) and involve future carers in discussions.

Available Tools

- **Programmes:** **Good 2 Go** (Toronto SickKids), **Ready Steady Go** (UK), **Got Transition** (US; six core elements, Fig. 65.2). **The best programme is the one that fits the unit.**

FIG. 65.2 SIX CORE ELEMENTS: A TIMELINE FOR ADOLESCENT AND YOUNG ADULT (AYA) TRANSITION FROM PAEDIATRIC TO ADULT HEALTH CARE

Element	Age (years)	Action
Policy/guide	12–14	Develop, discuss and share a transition policy
Tracking and monitoring	14–18	Track progress with a flow-sheet registry
Readiness	14–18	Assess self-care skills; educate on identified needs
Planning	14–18	Transition plan with a medical summary
Transfer of care	18–21	Transfer to an adult-centred practice
Transition complete	18–23	Confirm completion; seek patient and parent feedback

- **Pattern:** the three 14–18 elements run **in parallel**; only policy, transfer and completion are sequential.
- **Trap:** transfer (18–21) is not the endpoint. Transition is complete only when **integration is confirmed and feedback obtained** (by 23).

Concept — *Transition is a **process started by the paediatric team and completed in the adult setting**. It begins at about 12, is tracked through adolescence, and ends only when integration into adult care is confirmed, not at the last paediatric appointment.*

SYSTEMS AND MODELS OF CARE

Patient Population

ROUTES INTO AN ADOLESCENT UROLOGY SERVICE

Route	Status	Reason
1. Transition and referral from the paediatric service	Ideal	Prepared patient
2. Referral from paediatrics without transition	Suboptimal	Integration easier than route 4 because there is no concurrent acute need
3. Referral with a new problem arising in adolescence	Ideal	New problem, managed from the outset
4. Emergency referral after discharge from paediatric care	Most challenging; most prevalent route in some systems	Unequipped with skills or knowledge of adult care; insurance obstacles because of pre-existing conditions

- **Trap:** route 3 is "ideal" even though no transition took place, because the problem is new.

Potential Models for the Provision of Lifelong Care

MODELS OF LIFELONG CARE

Model	Advantages	Drawbacks
1a. Lifelong paediatric urologist in a combined paediatric/adult department	Absolute continuity; no transfer; one urologist follows results throughout life; suits limited life expectancy (some paediatric cancers)	Patient outlives the provider's career; patient not equipped to navigate other adult specialties; a purely paediatric environment is unhealthy for the patient and surrounding paediatric patients
1b. Paediatric urologist integrated with a separate adult department	Continuity plus collaboration; mutual learning; co-location allows joint clinics and joint operating	Cost, geography/hospital, clinic privileging; depends on willingness to use surrounding expertise
2. Transfer to an adolescent/adult specialist at a separate institution	The specialist also coordinates the MDT; joint clinics; meeting the new team in the familiar paediatric setting	Communication across institutions; institutional barriers; family fear of leaving; needs careful preparation and mutual trust between teams
3. Direct transfer without a transition plan	None stated; the default where systems are slow to recognise the need	Patients invisible until an emergency; no deliberate preventive care; many centres lack adult expertise

- **Self-induced model 3:** patients move for college or work, find no local expertise, drop out and **present in crisis**.

Multidisciplinary Care Models: Which Diagnoses Need Multidisciplinary Care?

- **The need for multidisciplinary care does not define the need for lifelong care.** Single-system conditions (**bladder exstrophy, severe hypospadias**) may be managed by a urologist, but care depends on understanding their subtleties.
- **Adult urologists are more subspecialised,** so subspecialist support and **multidisciplinary surgery** are common.

TRANSIENT VS LONG-TERM MULTIDISCIPLINARY CARE

Transient	Long-term
Single-system condition; input for a specific episode or operation	Multisystem concerns
e.g. a joint reconstructive operation	Most complex: neurological disorders (SB) , involving orthopaedics, neurosurgery, physiotherapy, urology and colorectal surgery

- **Psychology** for any condition; **social workers and care coordinators** for housing, employment and benefits.
- **The backbone of the service is nursing**, the front line of crisis management and care coordination.

Concept — Lifelong need and multidisciplinary need are independent axes. A single-system condition can need lifelong specialist care, and the most complex MDT needs come from multisystem neurological disease such as SB.

KEY POINTS

- Preparation for transition should begin in **early adolescence**.
- These are complex patients; **many need multispecialty care**.
- **Collaboration between paediatric and adult providers** is vital.
- **Failed transition leads to negative health outcomes**.
- Transition websites and tools help a unit develop a good service.

DEVELOPMENTAL CONSIDERATIONS

- **Factors in the decision to start treatment:** provider advice, **underlying risk (e.g. general anaesthesia)**, culture and religion, ease of access, and family resources (financial, intellectual, emotional). The **urologist is often decisive**.
- **Developmental paediatricians are usually excluded** from timing decisions, although the emotional and psychological effects persist into adulthood. Urologists have therefore developed **crude criteria**. For example, **augmentation cystoplasty is withheld until the child can help with catheterisation and says they want to be dry**, a judgement left to surgeon and parents alone.
- **Adult relevance:** could surgery have waited for **consent**, and what is the risk of **medical post-traumatic stress disorder (PTSD)**?

Development

- **Before the age of assent, two considerations apply:** surgery can **affect development**, physiologically and/or psychologically; and its **impact differs with life stage**, so **earlier intervention may cause less** emotional or physical morbidity.
- **Weak evidence base:** accepted beliefs about the timing of genital surgery (hypospadias repair, circumcision) rest on **mid-20th-century psychoanalytic research**, mainly observational narrative case histories.

DEVELOPMENTAL STAGES INVOKED FOR SURGICAL TIMING

Age	Stage	Surgical implication
First year	Bonding with an attentive caregiver; reciprocal emotion and communication	Avoid hospitalisation and separation
Second year	Exploration guided by the bonded caregiver	—
18–30 months	Individuation : expressing own desires, interacting beyond the nuclear family	—
3–6 years	Phallic-oedipal period : heightened genital awareness and emotional response to genital injury; Freudian anxiety about genital mutilation rises until puberty	Avoid genital surgery

RECOMMENDED WINDOWS FOR GENITAL SURGERY

Source	Window 1	Window 2
Developmental theory (maternal attachment, genital awareness)	< 6 months (before awareness)	2–3 years
AAP with a panel of paediatric urologists, 1975	< 6 months	4th or 5th year of life

- **Both agree on < 6 months but differ on the second window; there is no universal agreement.**

ELECTIVE VS SEMI-ELECTIVE OR NON-ELECTIVE SURGERY

Elective (e.g. hypospadias repair)	Semi-elective or non-elective (e.g. bladder exstrophy closure)
In practice, timing is driven by starting school, parental anxiety about "fixing" the child, and surgeon availability more than by developmental theory	Proceed as soon as it is safe from an anaesthetic and physiological standpoint

Timing of Surgical Intervention

- **Near adolescence**, major surgery becomes difficult when it **depends on the adolescent's compliance**.
- **Self-management has four components: medical adherence, medical responsibility, self-care skills, condition knowledge**. Mastery commonly varies **into the third decade**, which raises the risk of a failed therapeutic relationship.
- **Emerging adults (18–25 years)** are legal adults and make their own medical decisions **unless specifically found to lack capacity**.
- **Informed consent**: the right to decide based on risks, benefits and alternatives. **Emerging adults may legally give or withhold consent**.
- **Capacity**: may be lacking because of cognitive or developmental disability or incapacitating illness, in which case a **parent or guardian** is involved. **More often, capacity is present but the processing skills for complex decisions are incomplete**.
- **Spina bifida**: usually **normal intelligence and literacy**, but more likely to have deficits in **working memory and executive function**, the domains critical to medical decision-making. A **trusted urologist may be the only reliable bridge** to safe decisions. Recruit extra resources, such as a nurse practitioner, to engage and consent these patients.

Medical PTSD and the Psychoemotional Impact of Chronic Care in Childhood

- **Definition**: medical (iatrogenic) PTSD follows **negative or traumatic experiences of medical care**. Chronic childhood illness also has psychoemotional effects on the child and family.

- **Effects on young-adult decision-making:**
 - **distrust** of providers and **delay in seeking care**;
 - **challenging recommendations** or seeking alternative treatments;
 - **avoiding responsibility** for their own health, having always had it managed by others;
 - anxiety, panic attacks and reliving past events, **especially around invasive procedures**.
- **Evidence: no English-language studies** report the incidence or impact of medical PTSD in **adults with congenital urological conditions**. **Congenital heart disease data** suggest that risk relates to **childhood surgical exposure**, and that **cognitive-behavioural therapy (CBT) with psychological support** alongside medical treatment may reduce the long-term impact.

Concept — *Childhood surgical timing has been governed by weak psychoanalytic theory and practical logistics rather than evidence. The adult urologist inherits the psychological consequences, so the recurring question is whether the decision could have waited for the patient's own consent.*

THE PRACTICALITIES: SETTING UP A SERVICE

Training

- **There is no defined training pathway.** Paediatric urologists believe they are best suited to this care; adult reconstructive urologists believe the same of themselves.
- Most providers train in **both paediatric and reconstructive urology**. Transitional sessions at urology conferences are increasing.

The Multidisciplinary Team

MULTIDISCIPLINARY TEAM (MDT) MEMBERSHIP

Core	Likely needed	Available as needed
Paediatric urologist	Colorectal surgeon	Fertility specialist
Transitional/adult urologist	Gynaecologist	Orthopaedic surgeon
Specialist nurses	Rehabilitation specialist	Endocrinologist
Complex-care coordinator	Social worker	Geneticist
Nephrologist	Transplant team	
Psychologist	Neurosurgeon; plastic surgeon	

- **Building relationships with key specialties is essential, major work** and determines the long-term success of the service.

Institutional and Political Support

- A dedicated clinic is **impossible without institutional commitment**. These patients need extra clinic time, resources and multispecialty visits, and are **underserved under fee-for-service** insurance.
- **Build the case** with a **business model** (patient numbers and types, projected resource needs). **Patient and family feedback** to the institution is compelling.

FIG. 65.3 TIPS FOR MANAGING YOUNG ADULTS TRANSITIONING TO ADULT-CENTRED CARE (Data from American Academy of Pediatrics, American Academy of Family Physicians, American College of Physicians, et al.: *Supporting the health care transition from adolescence to adulthood in the medical home*, Pediatrics, 2011)

Tip	Content
1. Decision-making responsibility	The patient is responsible for health-care decisions (starting or continuing therapy, timing and type of invasive procedures), supported by family, friends and the team
2. Self-management skills	Praise progress with appointments, prescriptions and web-based electronic medical record (EMR) use. Where the patient struggles, involve a social worker, nurse or family
3. Consent and confidentiality	Review the policies at the first visit, with family present and again after they leave , so the patient can say privately what may be discussed in front of family
4. How the practice operates	Adult practices rely on web-based EMR communication and have fewer ancillary and navigation staff . Explain how to get questions answered and appointments booked
5. Routine and after-hours care	Explain appropriate use of emergency and urgent care; widen access to nurse visits; standing orders for routine needs (e.g. urine cultures) to avoid unnecessary ED visits

- **Trap:** confidentiality is discussed **twice**, first with family present and then alone.

Barriers to Transition

BARRIERS TO TRANSITION AND THEIR SOLUTIONS

Level	Barrier	Solution
Patient/parent	Parents wedded to the paediatric provider ; patient disengaged; fear of unknown facilities, systems and providers ; loss of a beloved team	Build independence early: address the child directly, teach them to describe their diagnosis, involve them in decisions; readiness questionnaires to document progress
Paediatric provider	Many care for older patients in an adult setting, which often works well; the family's great trust in them	Humility: ask "Am I the best person to manage this patient?", and if not, involve the person who is
Adult provider	Biggest barrier: unfamiliarity with complex congenital conditions (not taught in residency); financial disincentives ; extra resource needs	Partnerships and networks; additional training or reliance on others' expertise
Institutional	Apathy towards a niche population; concern it will not be profitable	Host-institution commitment to infrastructure ; reimbursement support from philanthropic or other creative sources beyond payers
Health system	Cost and bureaucratic inertia	Patients and marketing tie the population to the system's mission

Concept — A transitional service rests on **relationships and institutional backing, not on a training pathway**. Each barrier is a failure of **familiarity** (patient with the adult system, adult urologist with the condition) or of **funding** (institution, health system).

Geographic Considerations

- **Volume drives quality**, so services are **centralised**, at the cost of **travel**. **Telehealth** helps, but in the US faces payment issues and **dual licensing across state lines**, and allows **no examination**.
- **Ideal: hub-and-spoke**, a full specialist hub networked with local providers.

Developing a Plan That Fits; Billing

- **No established model.** Drivers: the children's hospital's **upper age limit**, clinician pay, and where adult colleagues work. Needs protected clinic time, a **designated institutional lead**, a defined **access route** (call centre or care coordinator) and agreed **clinic privileges**.
- **Remuneration favours procedures** over complexity, coordination and time; complex perioperative risk is **not paid in proportion**. Many patients are **under-insured** (underemployment, **financial toxicity** of chronic illness): involve case managers, social workers and financial counsellors.

Measuring Success in Transition

- **The AHRQ report concedes there is no definition of effective transition.**
- **Proposed metric: one attended post-transfer appointment.** Measurable, but not necessarily a durable health outcome.
- **Failed transition** raises loss to follow-up and ED use; no study has yet shown a benefit of transition to overall health or quality of life.

KEY POINTS

- Train in both **paediatric and adult reconstructive urology**; build the team by direct contact.
- Managing patient and parent barriers is a core skill; **specialist nurses** are among the most vital members.
- Work with management to overcome systemic barriers.

PRIOR MALE URETHRAL RECONSTRUCTION

Hypospadias, epispadias, VACTERL association.

Surveillance

- **Adult hypospadias repair complications:** 10.1–37.5 % first repair; 27.5–63.6 % after a failed repair.
- **After paediatric repair, young adults report:** cosmetic dissatisfaction 17–36 % (about one-third), curvature ~12 %, erectile dysfunction ~19 %. **Proximal: about half revised by puberty** (median 16.5 years).
- **No agreed follow-up duration** (surgeon survey: 2 years, 5 years, puberty or adulthood, each roughly a quarter). **After proximal repair, follow into adult life.**
- **Bladder exstrophy–epispadias complex (BEEC): incompetent bladder neck, short phallus with dorsal chordee**, genital concerns limiting partners. **Follow up every 1–2 years.**

Persistent or Recurrent Chordee

- Recurs **years or decades later, more after proximal repair**. Proximal meta-analysis: ~25 % after dorsal plication alone vs 5 % after ventral lengthening.
- **BEEC: dorsal chordee almost universal**, with torsion or partial glans loss (Fig. 65.4: phallic insufficiency with torsion; classic dorsal chordee). **Full correction rarely possible (pubic diastasis, corporal disproportion): set realistic goals.**
- **Length is the main BEEC concern. Phalloplasty with staged prosthesis:** adequate satisfaction, ~50 % complications (mainly implant); most can orgasm.

Fertility

- **Mechanisms:** impaired semen flow; **retrograde urine and infection** into prostatic and ejaculatory ducts; short penis or ectopic meatus; **prior orchidopexy or orchiectomy** for undescended testis (UDT).
- **Many have never had fertility discussed.** Offer **one or two semen analyses**, plus **gonadotrophins** if testicular failure is suspected.

MALE FERTILITY: HYPOSPADIAS VS BEEC

	Hypospadias	BEEC
Paternity	Lower even without UDT (hazard ratio 0.79); more assisted reproduction (ART), most in proximal ; fewer partners also contribute	Of the 45 % who tried, 70.6 % succeeded , half with ART
Semen	Impaired mainly with coexisting UDT , with raised follicle-stimulating hormone (FSH) (testicular failure)	Retrograde ejaculation in nearly half; over half azoospermic → early fertility referral

Psychosocial and Emotional

- **Hypospadias:** psychological impact driven by **number of operations** and **final penile appearance**, not **meatal position**. Decisional regret **54.7 % mild**, **12 % moderate to strong**; higher with **distal** (vs proximal) disease and adult dissatisfaction with cosmesis or length.
- **BEEC:** psychosocial effects **more pervasive**, **worse in males**. But **health-related quality of life equals the general population** and **tracks urinary continence**.

Concept — After male genital reconstruction, distress is **cumulative**: the number of operations predicts it, not the original severity. Fertility is often never raised, so **ask explicitly**.

PRIOR FEMALE GENITAL RECONSTRUCTION (BEEC, CLOACA)

Surveillance

- **No society guidelines**, but follow **through puberty into adulthood**: genital malformations affect menarche, sexual function, fertility and pelvic health.
- After cloacal repair: 46 % void spontaneously, 23 % incontinent, haematometra 38 %, primary amenorrhoea 25 %.
- Cloaca: CKD in ~half, 17 % reach end-stage renal disease (ESRD); solitary kidney plus reflux is highest risk. Renal function and urine protein at least annually; pre-conception referral to high-risk obstetrics and nephrology.
- Müllerian and renal anomalies present as **dysmenorrhoea or pelvic pain** (Fig. 65.5: age 26, solitary kidney with a remnant ureter draining into the vaginal apex).

Sexual Function

- **BEEC:** dyspareunia in half; **one-third** meet criteria for dysfunction, yet **78 % satisfied**. Short vagina, introital stenosis; pain introital or cervical, worse with prolapse.
- Cloaca after vaginal mobilisation: worse; only 42 % sexually active.
- Screen all, with or without vaginoplasty; **early pelvic floor physiotherapy** for reactive vaginismus.

Fertility and Pregnancy

- **BEEC hypofertility:** 68 % of those trying conceived, often **after > 1 year or with ART**; only 21 % without ART in year 1. **Anorectal malformation (ARM): Müllerian anomalies in over half**, commonest uterus didelphys; **at least half** after complex vaginal reconstruction **need ART**.

- **BEEC pregnancy: loss ~20 %, fetal loss 5–7 %, more preterm birth. Intermittent self-catheterisation (ISC) fails as the uterus grows** (indwelling catheter or nephrostomy); **urinary tract infection (UTI) in one-third to two-thirds** with supravescical reconstruction. **Wide pelvis → breech.**
- **Vaginal delivery after perineal reconstruction:** high rates of **third- and fourth-degree tears.**
- **Reflux nephropathy in pregnancy:** hypertension **odds ratio (OR) 5.55**, **pre-eclampsia OR 6.04**, miscarriage **11–23 %**. **Permanent renal loss risk:** renal scars, hypertension or moderate-to-severe proteinuria; co-manage with nephrology.

DELIVERY ROUTE

Group	Recommendation	Reason
BEEC	Planned caesarean , with gynaecology and urology	Fetal death mostly with emergency caesarean ; breech; prolapse
Prior cloacal repair, vaginal interposition or urinary reconstruction		Vaginal, rectal and anal injury; bladder neck or rectal fistula

Pelvic Organ Prolapse (POP)

- **Mechanism:** compromised **levator ani**, **wider and longer hiatus**, **deficient cardinal ligament.**
- **18–52 %**, from **late adolescence**, with or without pregnancy; **acute postpartum prolapse in half.**
- **In pregnancy: bed rest and pessary.** Surgical series are small, recurrence high, **no consensus: experienced prolapse surgeons only.**

KEY POINTS

- Female ARM and BEEC often include genital malformations.
- Co-manage pregnancy with **maternal-fetal medicine**, **urology ± nephrology.**
- **Offer elective caesarean** after bladder reconstruction.

Concept — *In females with BEEC or cloaca, pelvic anatomy dictates adult care: a **deficient pelvic floor** gives prolapse, **Müllerian and renal anomalies** give pain and renal risk, and a **reconstructed outlet** makes **planned caesarean** the safe route.*

BLADDER RECONSTRUCTION AND URINARY DIVERSION

Augmentation or Substitution Cystoplasty

- **Late stones, contraction, sacculation.** The segment used is often unknown; useful, rarely essential. A **cystogram** shows emptying, shape, urethra and bladder neck.
- **Ileum: annual vitamin B₁₂ from 5–10 years after augmentation;** deficiency is irreversible and **masked in SB.** Check **bicarbonate and chloride** for **hyperchloraemic acidosis** (ileum, colon).
- **Urine pregnancy tests: 57–75 % false positive.** Use **serum human chorionic gonadotrophin (hCG)** before teratogenic drugs or procedures.
- **Perforation 0.8–13 %** (overdistension, anastomotic ischaemia), linked to **substance misuse and missed catheterisation;** **mortality up to 25 % if diagnosis is delayed.** Friable tissue: **wide drainage with large-bore catheters.** With a **ventriculoperitoneal shunt (VPS)**, involve neurosurgery and infectious diseases.

Continent Diversion

- **Appendicovesicostomy beats Monti–Yang: ~50 % lower revision.** **Skin-level stenosis commonest;** fascial or valve stenosis presents with **perforation.**

- **Cannot catheterise:** usually **overdistension kinking the channel**. Drain per urethra or with a **floppy-tipped catheter**; else **percutaneous aspiration** (16–18 G), best through the **old suprapubic tube site**.
- **False passage:** endoscopic or fluoroscopic **dilatation over a wire**, catheter for a few days; then revision.
- **Chronic difficulty** (dexterity, obesity, cognition, support): consider **incontinent diversion or suprapubic tube**.

Incontinent Diversion (Conduits)

Problem	Features	Management
1. Stomal (up to 34.4 %)	Early stenosis ; later mesenteric tension and ischaemia. Parastomal hernia	Local repair almost always recurs (up to 56 %): keyhole or Sugarbaker mesh
2. Conduit stricture or dilatation	Lengthens, becomes tortuous; UTI, hydronephrosis	Loopogram with drainage films
3. Ureteroileal stricture (~10 %)	Hydronephrosis, loin pain, one-sided pyelonephritis	Endoscopic first (open repair morbid, often fails)

Bladder Neck Reconstruction (BNR)

- Creates a **fixed outlet obstruction**. BEEC after BNR: **59 % void per urethra; 64 % of these continent**.
- **Annual renal and bladder ultrasound (RBUS) and serum chemistry for secondary reflux or high-pressure storage**; if abnormal, **urodynamics ± renography**. Stones: ensure emptying, then **irrigation**.

KEY POINTS

- Know reconstruction complications; **review every 1–2 years with RBUS and bloods**.
- Let wheelchair use and hand function guide operative choices.

Concept — *Bowel in the urinary tract brings a **fixed list of lifelong risks**: B_{12} , acid-base, false-positive urine hCG, perforation, stones, channel stenosis. Surveillance is the **annual checklist** of that list.*

SURVEILLANCE RECOMMENDATIONS

Nephrologic Surveillance

- **Risk depends on** whether bladder and kidneys are both affected, childhood insults, and comorbidity (diabetes, hypertension).
- **Congenital obstructive nephropathy: highest risk. Puberty is high-risk** in congenital anomalies of the kidney and urinary tract (CAKUT). **Posterior urethral valves (PUV) progress in late adolescence, just when patients disengage**.
- **SB: CKD in one-quarter to one-half**; only **62 %** of registry patients had creatinine and RBUS within 2 years.

RENAL SURVEILLANCE IN SPINA BIFIDA

Spina Bifida Association (SBA), age 6 to adulthood	Authors' additions
Annual urinalysis, creatinine, RBUS; non-creatinine glomerular filtration rate (GFR) if "low muscle mass"; chemistry and B_{12} after reconstruction; no routine urodynamics beyond childhood	Annual cystatin C if non-weight-bearing or sarcopenic . Urodynamics for falling estimated GFR (eGFR), hydronephrosis, changed continence or escalating UTI

- **Creatinine** depends on muscle mass and **overestimates GFR in non-ambulant SB** (thoracic worse than sacral). **Cystatin C** is **independent of muscle mass**.
- **Trap:** "UTI" symptoms may be a **changing bladder**, not infection.

- **Acidosis:** usually subclinical; if true, **oral sodium bicarbonate**, or bone and kidneys suffer.
- **CKD (National Kidney Foundation):** kidney damage or **eGFR < 60** for **≥ 3 months**. **Avoid NSAIDs**, lithium, bisphosphonates, gabapentin; limit contrast; treat hypertension; **angiotensin-converting enzyme (ACE) inhibitor or angiotensin receptor blocker (ARB) first line** for proteinuria.

WHEN TO REFER TO NEPHROLOGY

First, optimise urologically	Exclude storage or emptying problems, obstruction, stones
eGFR	≤ 30 (CKD 4–5), or sustained loss ≥ 5 per year (mL/min/1.73 m ²)
Proteinuria	Albumin/creatinine ≥ 300 mg/g (≥ 30 mg/mmol) or ≥ 300 mg/24 h
Other	Unexplained haematuria after urological work-up; secondary hyperparathyroidism, anion-gap acidosis or non-iron-deficiency anaemia (recurrent stones); refractory hypertension; potassium abnormalities

Imaging

- **Annual ultrasound** with any upper-tract dilatation or lower-tract dysfunction or reconstruction. Increasing dilatation: **video urodynamics** and/or **MAG3 (mercaptoacetyltriglycine) renogram**.
- **"Tempered" MAG3: catheter in place** if the bladder or VUR may cause the dilatation; not routine in adults, so request it.

URINARY TRACT INFECTIONS

- **Bacteriuria in 50–100 % on intermittent catheterisation;** treat only **symptomatic** bacteriuria. Learn each patient's own signs.
- **Prevention: irrigation ≥ 240 mL**, good fluid intake, catheter technique and adherence.
- **Same organism repeatedly → uncleared source** (stone, diverticulum); **different organisms → reinoculation**. RBUS with post-void imaging.
- New symptoms: culture, more fluid and irrigation; **self-start antibiotics after culture** for rapid deteriorators. **> 4 infections a year → investigate**.

LONG-TERM RISKS

Malignancy

- **Congenital neurogenic bladder raises urothelial cancer risk**, with or without bowel; also exstrophy, immunosuppression, inflammation, stones, **haematuria–dysuria syndrome**.
- **Ileocystoplasty ~1.2 %; gastrocystoplasty 2.8–10.3 % (2–10-fold)**. Mixed histology; **latency 10–20 years; nearly always fatal**.
- **Annual screening cystoscopy is not cost-effective or accurate**. **Cystoscopy for symptoms** (pain, haematuria, frequent UTI) or a **filling defect**.

Stones

- **Up to 40 %**, more in **substitution**; **2 %** with **dependent, complete drainage**. **Daily high-volume flushes** reduce them. Many adults use **10–12 Fr** catheters, too small to irrigate.

RESERVOIR STONES BY SIZE

< 1.5 cm	Fragment and extract via the Mitrofanoff channel
Slightly larger	Ultrasound-guided percutaneous , protecting the channel
> 2 cm or multiple	Open cystolitholapaxy , without fragmentation

- **Leave no fragments:** they cause early recurrence.

CHANNELS AND STOMAS

- **Mitrofanoff revision 30–50 %**, mainly **stenosis** (skin level) and leakage; weight gain and tortuosity add difficulty. Examine in the patient's **usual position** (e.g. wheelchair).
- **Skin-level stenosis:** indwelling catheter for a few days or a **night-time "L-stent"** (Fig. 65.6: the patient's catheter, bent and taped into a Mitrofanoff or Malone antegrade continence enema (MACE) channel to dilate it). Then **excision with advancement flap or oral mucosa**.
- **Conduit stoma complications 18–35 %**. **Digital examination** localises stenosis relative to fascia. Retention → **loopogram**; hernia planning → **CT**; nephrostomy or equivocal loopogram → **antegrade nephrostogram**.
- **Appliance leakage: stoma nurses first**; they may avoid surgery and **mark the revision site**.
- **Metabolic: acidosis (alkalosis with gastric segments)** and B₁₂: annual bloods.
- **Abdominal wall:** SB, BEEC and **Eagle–Barrett** predispose to hernias; high recurrence, so involve wall specialists. In multiply operated patients, **almost any procedure carries a 91.5 % complication risk** (mostly minor).
- **Change in reservoir function: shorter dry interval, new enuresis, or new continence** (hostile bladder behind a tighter outlet) → **urodynamics ± voiding cystourethrogram (VCUG)**. **No routine adult urodynamics**; use for upper-tract change, incontinence or **hypercontinence**, escalating UTI.

KEY POINTS

- **Treat symptomatic bacteriuria early.** Collaborate on complex problems.
- **Changes in continence require investigation.**

Concept — *In a reconstructed tract, **change is the signal**. A new pattern of continence, infection or dilatation prompts investigation; stable bacteriuria does not.*

PROGRESSION TO ESRD AND RENAL REPLACEMENT

- **CAKUT: 34–43 % of childhood ESRD. PUV: 23 % in ESRD by median age 31.**
- **SB patients are transplanted less often** ($P < 0.001$) despite **comparable survival** on renal replacement therapy (RRT): complexity, transplant teams' uncertainty about the bladder, patients "worn out".
- **Psychosocial:** patients lost in childhood **reappear with severe renal failure**; family **guilt, blame and denial**; **more depression** (a driver of non-adherence); carer distress.

FIG. 65.7 CHALLENGES IN EMERGING ADULTS WITH ESRD (Modified from Diaz-Gonzalez de Ferris ME, Blood Purif, 2011)

1. Brain development	Prefrontal executive function matures by the mid-to-late 20s ; lower in SB
2. Socialisation, self-esteem	Lower employment and education
3. Non-adherence	Graft failure and progression to RRT peak at this age
4. Independence	RRT on top of incomplete transition
5–6. Age at diagnosis; cause	Time to adjust; heavy self-care and visible disability
7. Appearance	Short stature, vascular access, steroid facies
8. Vascular disease	Early calcification , a leading cause of death
9. Cognition	Longer ESRD, greater deficit

Management of Polyuria

- Most marked in **PUV**: up to **one-quarter lose function after puberty**; up to **75 %** have **instability, myogenic failure or poor compliance** after adolescence; polyuria may produce a "**VURD**" kidney (VUR with renal dysplasia).
- **Double voiding; mainstay: overnight indwelling catheter. Native nephrectomy is discouraged**: residual urine eases dialysis and blood pressure. Overactivity: behavioural → anticholinergic or β_3 agonist → botulinum toxin → augmentation.
- **Before any surgery, plan future graft sites** so incisions, ports and channels leave room.
- **Bladder re-evaluation at listing is mandatory. Small, poorly compliant → augment before graft. Myogenic failure → reliable emptying (Mitrofanoff): inconsistent catheterisation causes graft failure.**

Native Kidneys and Peritoneal Dialysis (PD)

- **Native nephrectomy indications**: recurrent pyelonephritis, recurrent stones, malignant hypertension, chronic flank pain (reflux), severe nephrotic syndrome. Pyelonephritis often starts **after transplant** (Fig. 65.8: ileal conduit, native **emphysematous pyelitis** with reflux into the native ureter; nephroureterectomy). **Massive reflux: remove the whole collecting system**, leaving no **orphan stump**.
- **PD relative contraindications: dense adhesions, VPS**. Channels and stomas raise infection concern (small SB series: no excess, underpowered). **VPS problem on PD → non-peritoneal shunt drainage. Poor sensation delays peritonitis diagnosis**: act early on cloudy effluent.

Transplant Considerations

- **Transplant is the preferred RRT**. Neuropathic bladder work-up: **cystoscopy and video urodynamics** (or urodynamics plus VCUG). **Patient must show willingness to do bladder self-care**.
- **Pyelonephritis 0.49 vs 0.31–0.44 episodes per patient-year; graft survival 92.3 % at 5 years**, similar to general recipients.

VASCULAR AND POSITIONAL PLANNING

VACTERL	Anomalous vessels → MR or CT angiography
BEEC	No crossover flow to the corpora: hypogastric artery use risks hemi-penile erectile dysfunction
Stomas and channels	Mostly midline or right → first graft in the left pelvis

- **Unexplained rising creatinine or cystatin C → urodynamics and imaging;** the urologist partners with the transplant team throughout.

Concept — *The bladder decides the graft.* *Plan the graft site before reconstruction, reassess at listing, and secure reliable emptying.*